



TRAVELOS — CORPORATE TRAVEL MANAGEMENT

End-to-End System Flow, Technical Wireframes & Architecture Blueprint

ANGULAR
16+

SPRING BOOT
3.2

JWT +
RBAC

H2 +
JPA

STAKEHOLDERS

1

Traveling Employee

Raise · Track · Upload · Close

2

Approving Manager

Review · Remark · Decision

3

Travel Desk Team

Bookings · Timeline · Assets

4

Financial Team

Audit Claims · Credit Sal

5

Risk & Admin

Policy · SOS · Governance

SYSTEM FLOW (END-TO-END LIFECYCLE)

Phase 1: Landing & Login

Auth Module

ID: EMP001 / EMP002

PW:

Log In → JWT Bearer

Role Passcodes

- Manager: 111111
- Travel Desk: 222222
- Finance: 333333
- Admin: 444444

Phase 2: Employee Dashboard

Employee Hub

Profile Info

- Name: Arjun S.
- Dept: Engg
- Govt ID: Verified

Trip Registry

+ Raise Request

Active	Closed	Upcoming
TRIP-1024	TRIP-0982	TRIP-1055

Phase 3: Create Request

Form

- Project & Client: #8492 · Client #C-901
- Route: Mumbai → London (7 Days)
- Hardware Assets: ThinkPad X1 + Token

Flight Hotel Cab Luggage

Est: \$3,450

Submit → PENDING

Phase 4: Auto-Validation

5 Checks

- Emp Active
- Client Valid
- Company Policy Passed
- Project OK
- Budget OK

✓ PASS → MGR

✗ FAIL → REJECT

Phase 5: Manager Review

Manager

Pending: 3 | Approved: 14 | Rejected: 1

- Audit: Purpose & budget fit confirmed
- Assets: Hardware allocation signed off

[APPROVE]

[REJECT]

Phase 6: Travel Desk & Booking

Bookings

- Flight: PNR AI-204 (08:00 AM → 11:00 AM)
- Cab: DL-01-AB-1234 (Ramesh Kumar)
- Hotel: Grand Hyatt (Booking #HBK-90234)
- Assets: HW-Asset-8492 dispatched

Activate Itinerary → ACTIVE

Phase 7: Live Tracking & Proof

7 Milestones

1 Flight Board Seat 14A	2 Flight Land Term 2	3 Cab Pickup OTP 4829	4 Hotel In HBK-902	5 Hotel Out Pending	6 Return Fit Pending	7 Journey End Pending
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* Mandatory Proof Verification enforces live milestone progression.

Phase 8: Financial Settlement

Finance

- Upload Bills: Receipts for Meals, Local Cab, Emergency
- Audit: Accounts reviews claim validity & receipts
- Reimbursement: Approved amount CREDITED to salary

* Requires 'CREDITED' state before trip closure eligibility.

Phase 9: Trip Closure & Audit

4 Gates

- ① Journey Ended: Milestone 7 verified with proof
- ② Assets Returned: Hardware logged back in inventory
- ③ Expenses Settled: All claims in CREDITED state
- ④ End Date Passed: Current date exceeds trip return date

4/4 Gates Verified

[CLOSE TRIP] → CLOSED

KEY REST ENDPOINTS (24 APIS)

POST /api/trips PUT /trips/{id}/approve
POST /itinerary/{id}/timeline PATCH /trips/{id}/milestone
PUT /expenses/{id}/credit POST /trips/{id}/close

DATABASE ENTITIES (8 TABLES)

User TripRequest TripItinerary TripMilestones
CheckListTimeline Expense

SECURITY & INTEGRITY

JWT 24h Bearer Role-Based Guard
H2 File Persistence Proof Verification

Frontend: Angular 16+ · Glassmorphism · RxJS Reactive Store

Backend: Spring Boot 3.2 · Java 17+ · Spring Security

Database: H2 File-based DB · Hibernate JPA

Specs: 24 REST Endpoints · 8 Entities · 7 Milestones